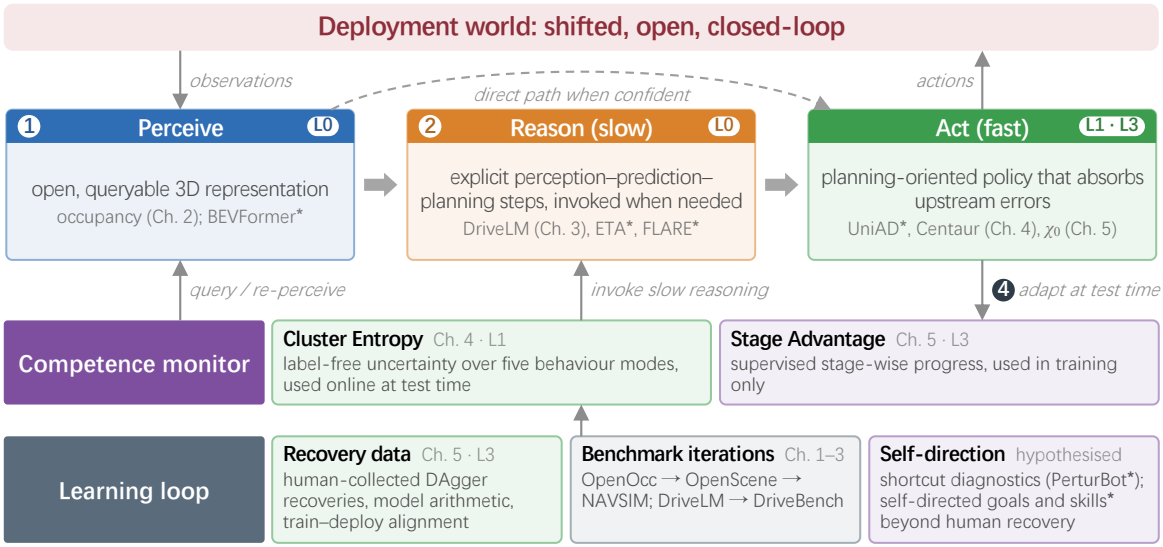

Deployment world: shifted, open, closed-loop
observations
direct path when confident
actions
1 Perceive L0
open, queryable 3D representation
occupancy (Ch. 2); BEVFormer*
2 Reason (slow) L0
explicit perception–prediction–planning steps, invoked when needed
DriveLM (Ch. 3), ETA*, FLARE*
Act (fast) L1 · L3
planning-oriented policy that absorbs upstream errors
UniAD*, Centaur (Ch. 4), χ_0 (Ch. 5)
query / re-perceive
invoke slow reasoning
4 adapt at test time
Competence monitor
Cluster Entropy Ch. 4 · L1
label-free uncertainty over five behaviour modes, used online at test time
Stage Advantage Ch. 5 · L3
supervised stage-wise progress, used in training only
Learning loop
Recovery data Ch. 5 · L3
human-collected DAgger recoveries, model arithmetic, train–deploy alignment
Benchmark iterations Ch. 1–3
OpenOcc → OpenScene → NAVSIM; DriveLM → DriveBench
Self-direction hypothesised
shortcut diagnostics (PerturBot*); self-directed goals and skills* beyond human recovery